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Pushing the Limit of Formation Pressures Data Acquisition in HPHT, Sour Tight Carbonates of N. Kuwait Jurassic Gas with Differential Depletion Challenges and Opening New WRM, Development Frontiers

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Abstract

Better understanding of current formation pressure in differentially depleted carbonate and sour reservoirs of North Kuwait's HPHT Jurassic Asset (~12,000 psi, ~285 °F) is critical for Well & Reservoir Management (WRM) and further field development. These tight, heterogeneous reservoirs (porosity <10 %) exhibit significant depletion and pressure variation (~3000-11000 psi) in several zones. Recent shifts in KOC's field development strategy include horizontal and underbalanced coiled tubing drilling (UBCTD).

Due to large pressure variation, precise formation pressure data is essential for well placement, managing stress-related risks with geomechanical models, and ultimately enabling targeted perforation and development strategies.

Traditionally, in highly deviated wells, pressure measurements are often excluded due to high risk of differential and mechanical sticking. Furthermore, formation tightness poses challenges in achieving stable pressure readings. To address these limitations and optimize rig time, a fast wireline pretest tool (PressureXpress) with dynamic pretest control was deployed successfully. This study further integrates pressure data from multiple sources - Production Logging (PL), Static Bottom-Hole Pressure (SBHP) and Formation Tester (FT) to enhance reservoir understanding. The integrated dataset supports WRM decisions and field development planning by enabling accurate depletion mapping, compartmentalization analysis, and improved well placement strategies in complex HPHT environments.

In tight, complex reservoirs such as North Kuwait's Middle Marrat formation, acquiring reliable zonal contribution data using multirate production logging (PLT) is highly challenging due to low flow rates and crossflow effects. Reliable PLT data is therefore difficult to obtain, making pressure measurements in open hole essential for evaluating reservoir behavior. This paper also demonstrates that a formation tester rated for 8,000 psi differential pressure was operated near its upper design limits without operational anomalies. Multiple pressure points were acquired across the reservoir, revealing distinct pressure regimes and significant depletion in certain intervals. Successfully acquired formation pressures highlighted current zonal pressures ranging from ~3,000 to 11,000 psi, caused by the tight limestone matrix and anhydrite

acting as vertical flow barriers. Interval Pressure Transient Testing (IPTT) performed on the buildup data enabled estimation of permeability across zones. Reservoir pressure and permeability profiling played a critical role in bridging the gap between static and dynamic in-place volumes, identifying development targets, optimizing zonal coverage, and supporting effective field development planning in these tight HPHT carbonate reservoirs.

This paper contributes novel insights by demonstrating how accurate formation pressure measurements are not only critical for dynamic reservoir characterization but also instrumental in planning advanced operational strategies such as Coiled Tubing Drilling (CTD) and stimulation. The integration of pressure data into WRM workflows further enhances development planning by identifying unlocked reservoir potential. As a result, this work not only reinforces the value of pressure data but also adds a practical framework for its application in improving field development efficiency in complex reservoirs.

North Kuwait Jurassic Gas Asset - Field Summary

The North Kuwait Jurassic Gas (NKJG) Field targets carbonate reservoirs that contain near-critical gas and volatile oil, with primary production from the Middle Marrat multi-flow units. These reservoirs present considerable challenges due to complex rock and fluid properties, as well as variable reservoir quality. Production operations are difficult both upstream and downstream, with wells typically reaching measured depths (MD) of 14,000 to 18,000 ft. The produced fluids are sour, containing 15% H₂S and 6% CO₂, and exhibit a wide range of reservoir pressures—from severely depleted (~3,000 psia) to near-virgin conditions (~11,000 psia)—with an average reservoir temperature of 275 °F. A location map of the North Kuwait Jurassic Gas (NKJG) Fields is shown in the figure below.

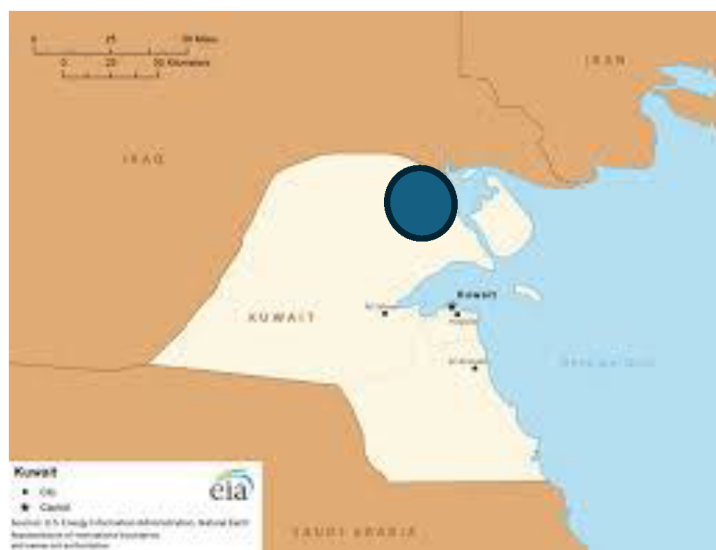
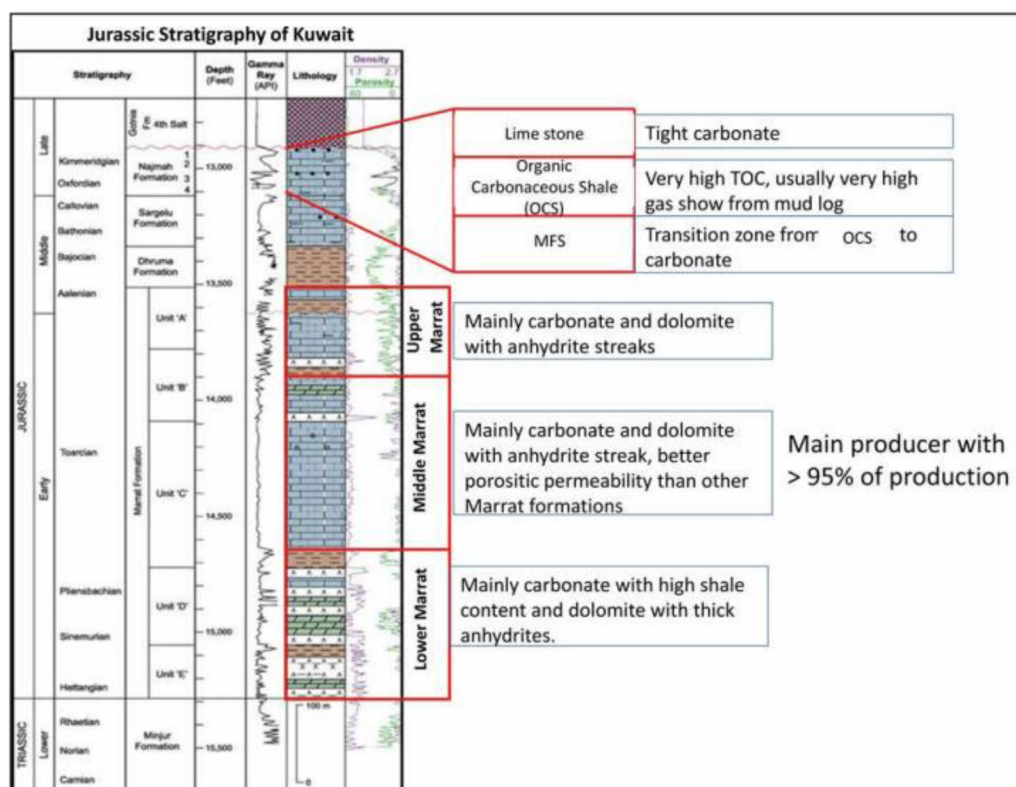


Figure 1—Map Location of NKJG Fields

The primary producing interval is a tight limestone (Calcite >90%) characterized by very low porosity and matrix permeability, interbedded with dolomitic streaks that offer better reservoir quality. Pressure Transient Analysis (PTA) conducted across various wells confirms dual-porosity behavior, where the matrix acts primarily as storage, with average matrix permeability below 1 mD.

Production began in 2008, with more than ~300 wells drilled to date in the Marrat reservoirs, producing predominantly under depletion- drive mechanisms. While most wells continue to flow naturally, liquid loading is increasingly evident, primarily due to ongoing pressure depletion and high surface backpressure caused by limitations in medium-pressure separation capacity at the high-pressure inlet facilities.

Due to the reservoir's tight nature, significant permeability variation across flow units, and the dual-permeability system (from matrix and natural fractures), well productivity is highly dependent on effective stimulation treatments. These treatments aim to enhance reservoir contact and activate natural fracture systems where present. The figure below illustrates the Kuwait Jurassic Stratigraphy (M. Abdel-Basset *et al.*, 2022).



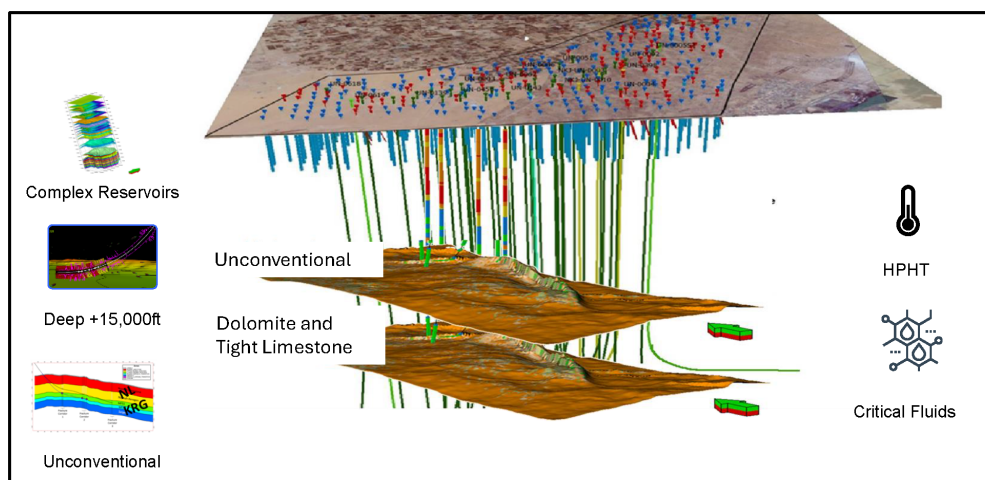


Figure 3—Schematic Showing NKJG Reservoir Overview

These are HPHT and critically sour tight carbonate reservoirs with pervious dolomite and very tight/impervious limestone matrix and anhydrite vertical flow barriers. The magnitude of differential pressure across different flow zones increases with production, resulting in drilling complications such as stuck pipe and borehole stability issues.

Better understanding of current formation pressures in differentially depleted carbonate and sour reservoirs of North Kuwait's HPHT Jurassic Asset (~12,000 psi, ~285 °F) is critical for WRM and further field development. These tight, heterogeneous reservoirs (porosity <10%) exhibit significant depletion and pressure variation (~3,000-11,000 psi) in several zones. Recent shifts in KOC's field development strategy include horizontal and CTD drilling with aggressive stimulation. Due to large pressure variation, precise formation pressure data are essential for well placement, managing stress-related risks with geomechanical models, and ultimately enabling targeted perforation and development strategies.

The plot below summarizes the field development strategy and its implementation. As stated, the FDP focuses on meeting the objective of the primary value driver, which is maximizing gas recovery. Hence, it is expected that once reservoir pressure falls below the saturation pressure, solution gas will evolve from the oil, leading to a steady rise in the gas-oil ratio (GOR) with time. Reservoir pressure decline relative to produced gas is expected to slow down once gas is liberated, and the same behavior can be seen in the figure below.

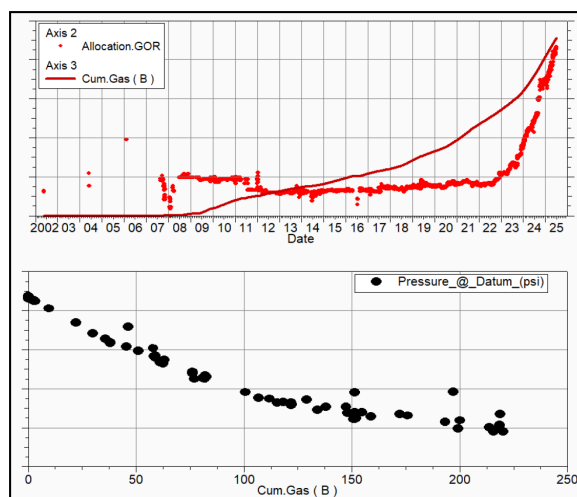


Figure 4—Production Pressure Performance Plot Exhibiting the FDP Strategy

Critical Need for Formation Pressure Acquisition in Infill Wells

Collecting formation pressure data during infill drilling is essential for making informed decisions in WRM and field development planning. This information becomes even more critical under KOC's evolving development strategy, which emphasizes horizontal wells, aggressive stimulation, and Coiled Tubing Drilling (CTD) in tight, low-permeability carbonate formations. In this context, formation pressure acquisition is no longer optional—it is a strategic necessity to unlock remaining reserves and extend field life in a technically and operationally challenging environment. The summary below describes the importance of gathering formation pressure data.



Overcoming Pressure Measurement Challenges in Low Mobility Formations

Accurately measuring formation pressure in low-mobility Jurassic formations presents significant technical challenges. In these tight reservoirs (0.05-1 mD/cP mobility), acquiring reliable pressure data with conventional formation testers is difficult. Many pretests in such intervals result in responses categorized as *Dry*, *Tight*, or affected by *Supercharging*.

This limitation arises from the fact that conventional hydraulic pretest mechanisms are not designed to handle the ultra-low fluid inflow rates associated with tight rock. These tools typically apply pretest drawdown rates that are too aggressive relative to the formation's ability to deliver fluid. As a result, the

entire drawdown phase is often dominated by flowline decompression. This pressure drop is caused by expansion of the compressible fluid already in the tool, with little or no contribution from the formation itself. In such cases, formation fluid does not enter the probe quickly enough, and the tool cannot record a valid pressure response or a meaningful buildup.

Historically, formation pressure testing in this field has been cancelled most of the time due to operational risk. This has led to a lack of proper reservoir fluid/productivity characterization and implied high uncertainties. Low permeability requires longer stationary times.

In addition, the high differential pressures associated with reservoir depletion must be managed carefully to avoid the risk of the tool getting stuck.

To overcome these challenges, PressureXpress™ (XPT) has been deployed successfully in these reservoirs. This electromechanically controlled tool provides three critical advantages:

- **Precise control over pretest volume and rate** through a **stepper motor-driven piston**, as opposed to the less controllable hydraulic systems in conventional tools. This enables drawdowns to occur at **very low and controlled flow rates**, which are essential for avoiding overdraw in tight formations and achieving pressure stabilization.
- **A significantly smaller flowline volume and tight mechanical construction**, which minimize the volume required for flowline decompression. This greatly reduces the time needed to reach valid drawdown conditions and shortens the overall test cycle, thereby reducing both **operational risk** and **exposure time in the well**.
- **Slim outer diameter (3-7/8")**, making the XPT ideal for deployment in 6-inch slim holes often encountered in Jurassic reservoirs. The reduced tool OD significantly lowers the risk of differential sticking, in contrast to standard-diameter tools which may face increased drag while deployment.

By leveraging these capabilities, the XPT was able to record high-quality formation pressures in zones where conventional tools might struggle.

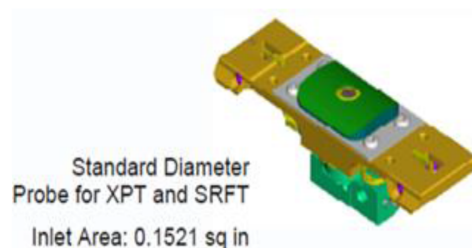


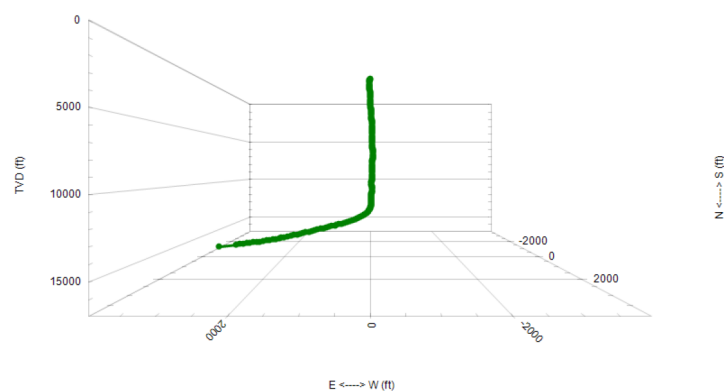
Figure 5—XPT Tool and the Probe

Case Study: Formation Pressure Acquisition Methodology & Challenges In WELL A

In this section, we present a case study from Well A in the North Kuwait Jurassic reservoir, focusing on the acquisition of accurate formation pressure and mobility measurements. The discussion includes the operational challenges encountered, the methodology adopted to obtain valid data, and the best practices that contributed to a successful wireline formation testing operation. The general details of the well are provided below.

Well Details

- Formation: Middle Marrat
- Wellbore OD: 6 in
- Total Depth (TD): 17750 ft
- Open Hole Interval: ~1020 ft
- Maximum Deviation: 61°
- Maximum Temperature: 282 °F



As shown in the open-hole log below, this interval indicates a predominantly carbonate reservoir composed mainly of limestone with intermittent dolomite streaks. The density-neutron log confirms the tight nature of the formation, with porosity developing in the dolomitized sections. Additionally, the resistivity log does not show significant contrast across some beds, suggesting poor fluid identification and low-permeability streaks. The borehole image log also indicates a relatively heterogeneous texture, with laminated features that may contribute to vertical and lateral compartmentalization. These characteristics, combined with a wellbore deviation of 61°, created a highly challenging environment for formation pressure acquisition. This complexity required highly controlled testing using XPT to ensure seal integrity, manage drawdown rates, and obtain valid data.

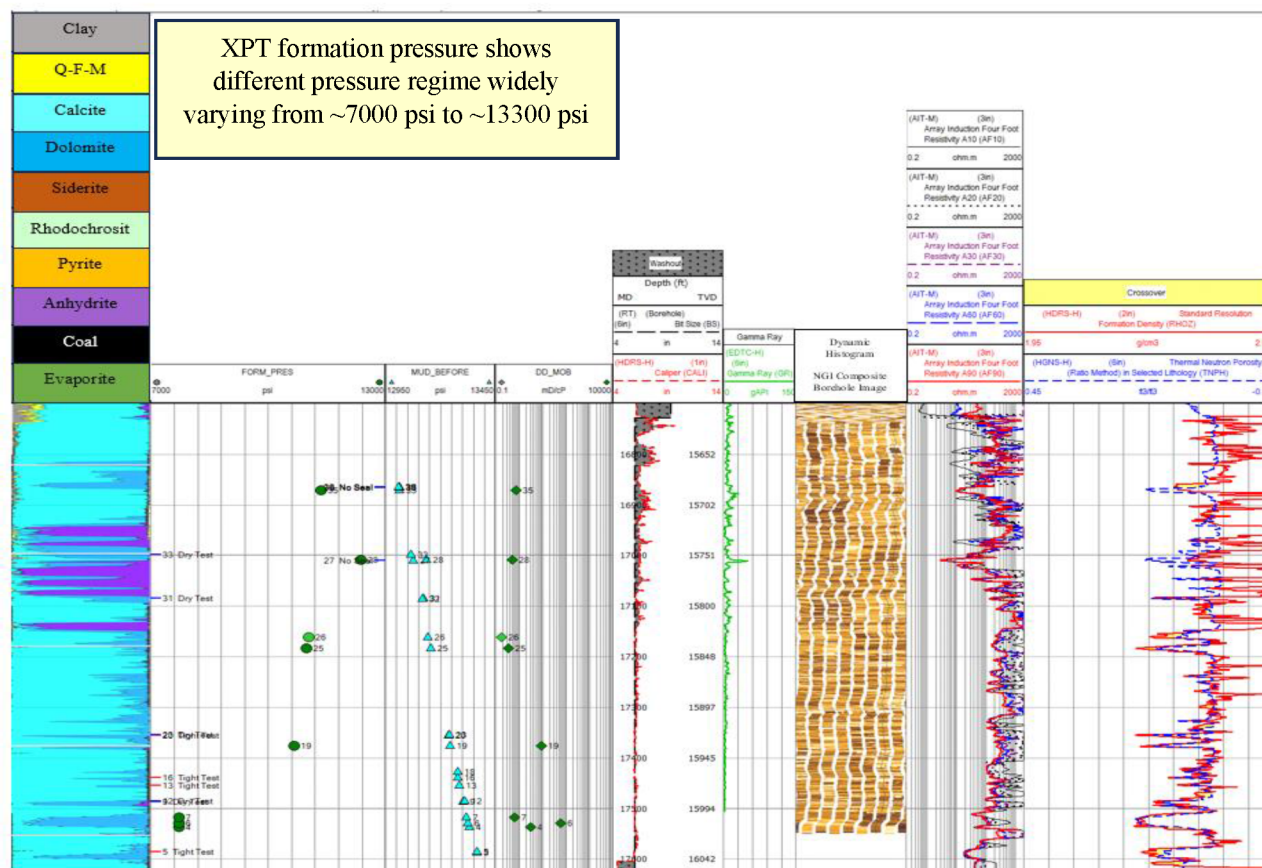


Figure 6—Open Hole Log Illustration

Methodology Followed for Safe and Reliable Pressure Acquisition

To ensure successful and safe pressure acquisition in a highly depleted environment, the following methodology was adopted:

Assessment of Overbalance and Risk Mitigation

Anticipated formation pressures were as low as ~3,000 psi, while hydrostatic pressure was around 13,300 psi, leading to a differential pressure of ~10,000 psi. Since this exceeded the XPT's rubber element differential pressure rating (8,000 psi), it was critical to minimize station time and avoid extended exposure at high overbalance.

Hole Condition Check: Stickiness Test

Before initiating pressure testing, a "stickiness test" was conducted by holding the tool stationary without probe extension for ~10 minutes. This helped assess borehole stability and mitigate the risk of tool sticking—especially critical in highly deviated wells.

Time-Limited Station Execution

A strict limit of 5 minutes per station (including set and retract) was enforced. The XPT tool's fast setting and retraction (typically <1 minute total) allowed enough time for a controlled, high-quality pretest within the remaining ~4 minutes.

Pretest Volume and Rate Control

Based on overbalance conditions of ~3-5 Kpsi, total compressibility was assumed to be $\sim 7 \times 10^{-6}$, and flowline decompression volume was estimated at ~1.2 to 2.4 cc.

The first pretest was executed at 3 cc and 0.1 cc/s to fully overcome flowline decompression and observe pressure drop below formation pressure, confirming reservoir contribution.

Once formation response was seen in the first pretest, the second pretest was conducted at lower rates of 0.05 cc/s. This helped achieve steady-state drawdown, minimized pressure disturbance, and improved measurement quality. Ensuring sufficient drawdown volume was key to validating that the observed pressure drop resulted from formation inflow rather than flowline decompression. In such cases, mobility calculations are considered reliable and representative.

Real-Time Drawdown Pressure Limit Adjustment

Drawdown pressure limits were monitored and adjusted in real time during testing. The initial limit was set conservatively. If tight/dry points were observed, the limit was lowered further to ensure that drawdown pressure went below the formation pressure.

A few examples are shown below in Figures for pressure points taken at different depths:

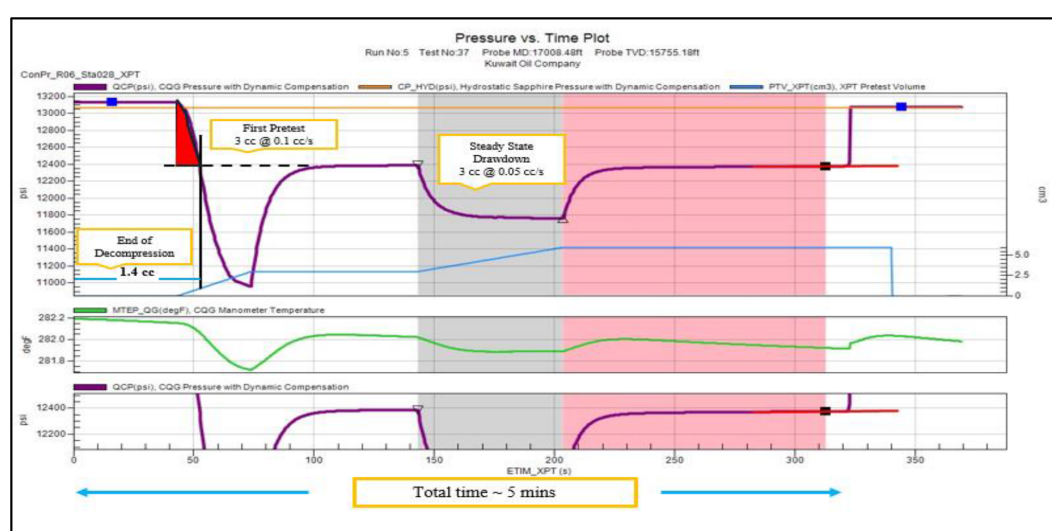


Figure 7—Pressure point example(Formation Pressure ~ 12375 psi, Mobility ~ 0.5 mD/cP)

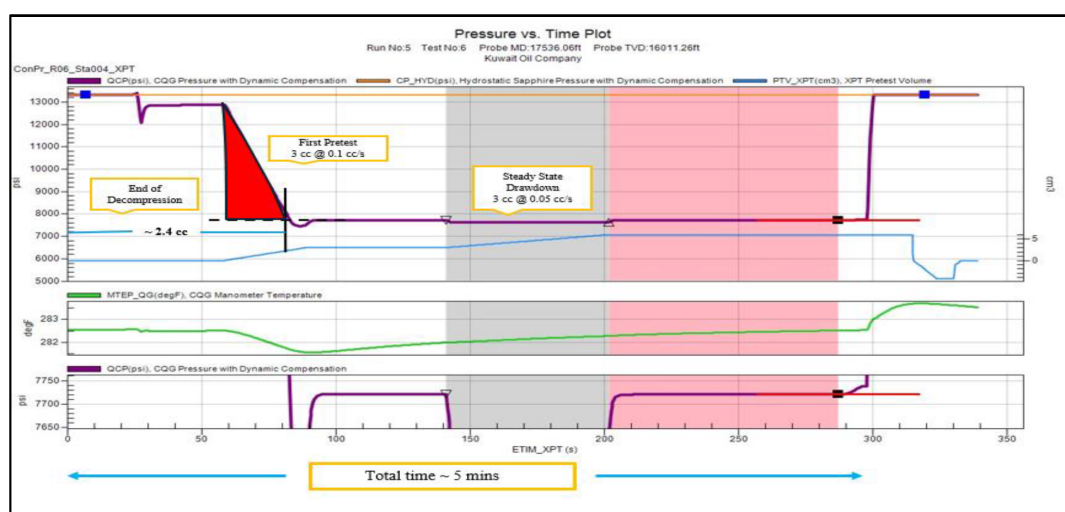


Figure 8—Pressure Point Example(Formation Pressure ~ 7720 psi, Mobility ~ 3 mD/cP)

The example shown in the figure below is from another Jurassic well and illustrates a similarly highly depleted environment, where formation pressure measurements were acquired using the XPT tool. The

measured reservoir pressure was approximately 2,000 psi, indicating significant depletion. Due to this, a high differential pressure was encountered across the probe rubber, which was effectively managed by applying a minimal pretest flow rate and setting the drawdown limit close to the estimated reservoir pressure. This approach helped ensure tool seal integrity throughout the job.

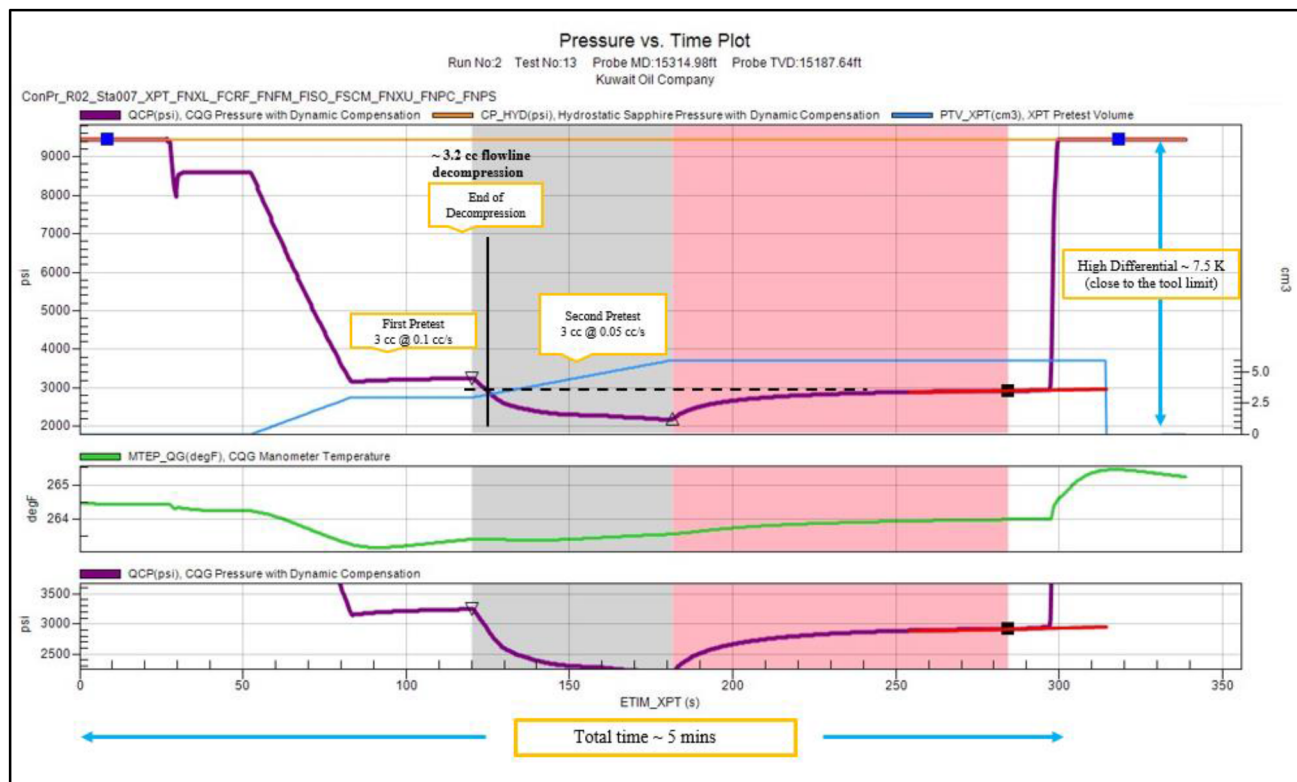


Figure 9—Pressure Point and Flow Regime Derivative Example(Formation Pressure ~ 2919 psi, Mobility ~ 0.3 mD/cP)

Observations, Results & Recommendations

A total of 17 pressure test points were attempted, of which 8 yielded valid formation pressure measurements for Well A, as shown in the figure above. These successful data points were acquired within a relatively short operational window, underlining the efficiency of data acquisition and the value delivered to the client in minimal time.

The measured formation pressures ranged from approximately 7,000 psi in depleted zones to 12,300 psi in virgin reservoir intervals, clearly indicating the presence of **multiple pressure regimes**. This wide variation suggests **reservoir compartmentalization and barriers**, all of which are critical for planning stimulation and production strategies. Mobility values also varied between <1 mD/cP and ~50 mD/cP, which may be indicative of a highly permeable streak.

These results are highly consequential for stimulation planning, particularly for hydraulic fracturing or acidizing operations. The key considerations include:

1. Zonal Diversion Strategy
 - Due to the high contrast in mobility, a conventional stimulation approach may lead to fluid channeling into the high- mobility/fractured zones, leaving the tighter zones untreated.
 - Mechanical or chemical diverters should be deployed to isolate and sequentially stimulate zones with lower mobility to ensure uniform treatment coverage and maximize reservoir contact.
2. Customized Pumping Schedules

- The depleted low-pressure zones (~7,000 psi) will require careful pressure ramp-up to avoid premature fluid loss. Conversely, virgin reservoir pressure zones (~12,300 psi) will need higher treating pressures and possibly increased fluid volumes to achieve effective fracture propagation.

3. Fracture Propagation Modelling

- The variation in formation pressure and mobility must be incorporated into geomechanical and reservoir models to predict fracture geometry and height containment. The high-mobility zone should be modeled to assess its potential as a thief zone, which can lead to fluid and proppant loss.

Wrm and Sub-Surface Excellence for FDP

Due to wells being completed in different flow zones, routine SBHP or pressure buildup surveys cannot provide the granularity of reservoir pressure at the level of each flow unit. This information is critical for identifying undepleted zones, which could be potential candidates for zone transfer or selective layer development by horizontal wells placed in higher-pressure flow units.

The figure below shows a plot of SBHP with time. As can be seen, SBHP surveys alone may lead to the incorrect inference that the entire field and all zones are depleting uniformly, leaving limited or no opportunities for zone transfer or further field development. SBHP surveys measure an average pressure across the open zones, which likely equalizes within those zones.

In contrast, formation pressure measurements conducted in open hole during recent infill drilling — between currently producing and depleted wells—have highlighted significant vertical heterogeneity and the presence of less-depleted to undepleted flow zones. Although selective layers have experienced high depletion from an initial ~11,000 psi to ~2,500 psi, the open-hole formation pressure measurement tool successfully recorded both near-virgin and highly depleted pressures within the same borehole.

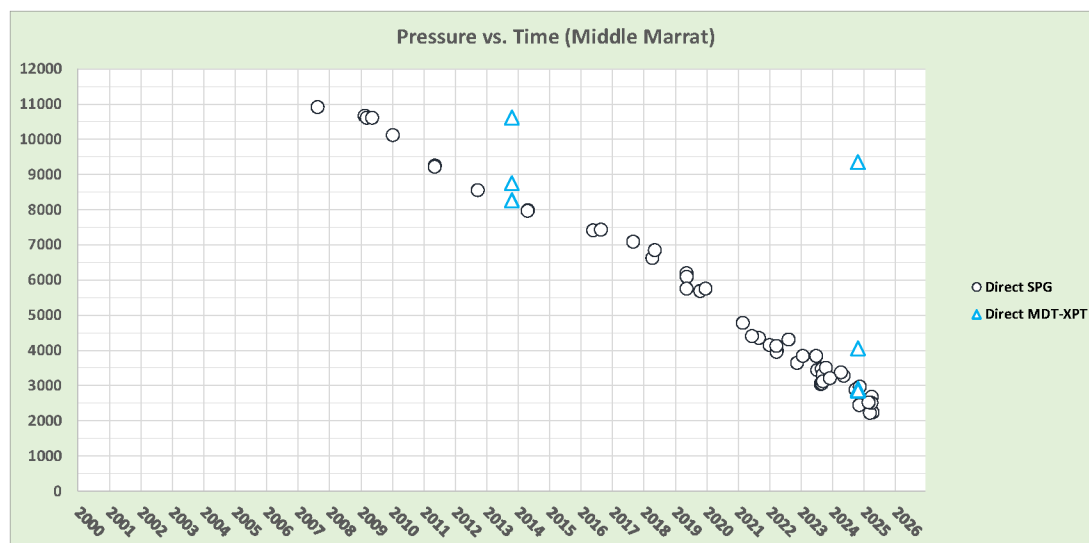


Figure 10—SBHP and Formation pressure measurement comparison

As shown in the table below, the wells are completed in multiple zones; however, not all zones are open in each well. This information is crucial for identifying non-depleted or slightly depleted zones, which can then be targeted through zone transfer or horizontal infill drilling focused on selected zones in different areas of the field.

Table 1—Current open flow zones and future opportunities (ZOI- zone of interest)

	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7
WELL 1	ZOI	PERF	PERF	PERF	PERF	ZOI	ZOI
WELL 2	PERF	ZOI	ZOI	ZOI	ZOI	ZOI	ZOI
WELL 3	PERF	PERF	PERF	ZOI	PERF	ZOI	ZOI
WELL 4	ZOI	ZOI	ZOI	ZOI	ZOI	ZOI	ZOI
WELL 5	PERF	ZOI	PERF	PERF	PERF	ZOI	ZOI
WELL 6	PERF	ZOI	PERF	ZOI	ZOI	ZOI	ZOI
WELL 7	PERF	ZOI	PERF	PERF	PERF	ZOI	ZOI
WELL 8	PERF	ZOI	ZOI	ZOI	ZOI	ZOI	ZOI
WELL 9	PERF	PERF	PERF	PERF	PERF	PERF	ZOI
WELL 10	ZOI	PERF	PERF	ZOI	ZOI	ZOI	ZOI

Conclusion

Optimizing Pretest Strategies for Low-Mobility Formations

Pressure measurement in formations with mobility below 1 mD/cP is often difficult, frequently resulting in poor-quality data such as supercharging effects, tight formation responses, dry tests, or micro-leakages. However, by employing improved pretest techniques, it is possible to significantly enhance data reliability. Optimizing the pretest volume and drawdown rate is an ongoing process throughout formation testing operations and must be adjusted dynamically.

Locate the remaining Gas (LTRG)

For optimum field development of an oil reservoir, a "locate the remaining oil" exercise is necessary to identify unswept areas and zones and accordingly modify the WRM and infill drilling campaign. Similarly, for a heterogeneous light oil/gas reservoir developed under depletion drive, a "locate the remaining gas" (LTRG) exercise is required to identify undepleted areas and adjust the WRM and further development plan accordingly. The success of formation pressure measurements in complex NKJG reservoirs has enabled the LTRG exercise and demonstrated the value of surveillance information for WRM and subsurface excellence in field development.

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Nomenclature

- ECD: Equivalent Circulating Density
- MPD: Managed Pressure Drilling
- CG: Connection Gas
- DPP: Dynamic Pore Pressure Testing
- XRD: X-Ray Diffraction
- MDT: Modular Formation Dynamics Tester
- XPT: PressureXpress Tool
- HPHT: High Pressure, High Temperature

FDP: Field Development Plan
LTRG: Locate the Remaining Gas
WRM: Well and Reservoir Management

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